

Prediction Horizon Definition Module

Module: PHDM — Prediction Horizon Definition Module

Parent Standard: POPS — Prediction Object & Purpose Standard

Architecture: PGA™ — Prediction Governance Architecture

Category: Governance & Enforcement

Subcategory: Prediction Governance

Governed Space: Prediction Horizon Definition

Primary Governed Object: Prediction Horizon

Module Type: Core Module

Canonical Language: UCL — Universal Canonical Language

Protection: MIP® — Methodological Intellectual Property

1. Canonical Definition

The Prediction Horizon Definition Module (PHDM) governs the establishment, specification, differentiation, coherence, and continuity of the temporal frame within which a Prediction Target is expected, assessed, or otherwise intended to apply.

PHDM establishes when the predictive proposition applies without determining whether the predicted outcome will occur, when it will actually occur, or how likely or feasible that occurrence may be.

Where time materially affects predictive meaning, the Prediction Horizon forms an essential part of the governed predictive frame.

2. Operational Role

PHDM is the fourth operational module within POPS.

It receives:

Identified Prediction Object + Defined Predictive Purpose + Specified Prediction Target

and establishes the Prediction Horizon applicable to that predictive proposition.

Its principal transition is:

Specified Prediction Target → Defined Prediction Horizon

PHDM prevents predictions applying to materially different temporal frames from being treated as though they were equivalent predictions.

3. Internal Module Flow

Prediction Object Identification → Predictive Purpose Definition → Prediction Target Specification → →
Prediction Horizon Definition → Predictive Boundary Establishment

Current Position: Prediction Horizon Definition



HDM produces the defined Prediction Horizon required for final establishment of the predictive frame by PBEM — Predictive Boundary Establishment Module .

4. Operational Space

PHDM operates within the Prediction Horizon Definition space.

It governs:

horizon identification; horizon specification; temporal reference; horizon type; horizon start; horizon end; prediction interval; temporal granularity; open and bounded horizons; horizon ambiguity; horizon differentiation; target–horizon coherence; horizon change; horizon continuity.

PHDM governs the temporal applicability of the prediction.

It does not govern actual outcome timing.

5. Operational Scope

PHDM applies wherever time materially affects the meaning, interpretation, feasibility, probability, monitoring, or eventual evaluation of a prediction.

A Prediction Horizon may be expressed through:

a specific future time;

a defined date; a bounded interval; a relative period; a recurring period; a defined operational cycle; an event-dependent temporal condition; a short-, medium-, or long-term predictive interval; another sufficiently identifiable temporal frame.

The appropriate horizon structure depends upon the predictive proposition.

6. Prediction Horizon Identification

PHDM must identify the temporal frame to which the Prediction Target applies.

The governing question is:

Within what temporal frame is the Prediction Target intended to be evaluated?

A target without an applicable temporal frame may remain materially ambiguous where timing affects its predictive meaning.

Prediction Target $\neq$ Prediction Horizon

7. Horizon Specification

A Prediction Horizon must be specified sufficiently to support subsequent predictive governance.

Where applicable, specification may establish:

temporal starting point; temporal endpoint; applicable interval; duration; recurrence; event-relative timing; temporal granularity.

The horizon must be sufficiently precise for its Predictive Purpose without introducing unsupported temporal precision.

Temporal Specificity $\neq$ Artificial Temporal Precision

8. Horizon Start

Where materially relevant, PHDM must establish when the Prediction Horizon begins.

The horizon start may correspond to:

prediction formation; prediction publication; a defined future timestamp; occurrence of another governed condition; beginning of an operational period; another explicitly established temporal reference.

These moments must not be assumed to be equivalent.

Prediction Creation Time $\neq$ Prediction Horizon Start

9. Horizon End

Where the prediction is temporally bounded, PHDM must establish the point or condition at which the Prediction Horizon ends.

The horizon end defines the temporal limit within which the Prediction Target is intended to apply.

Expiration of the Prediction Horizon does not itself determine whether the prediction was correct.

Outcome association and predictive performance belong to subsequent PGA™ governance.

Prediction Horizon End $\neq$ Outcome Determination

10. Point and Interval Horizons

PHDM must distinguish predictions concerning a particular temporal point from predictions concerning an interval.

A target expected:

at a specific time

is not automatically equivalent to a target expected:

at any time during an interval

or:

throughout an interval.

These horizon structures can materially change the predictive proposition.

Point Prediction Horizon $\neq$ Interval Prediction Horizon

11. Open and Bounded Horizons

A Prediction Horizon may be bounded or, where methodologically legitimate, open-ended.

An open horizon must not be used to create indefinite apparent validity where the predictive proposition requires meaningful temporal limitation.

Where the target can eventually occur at any unspecified future point, PHDM must preserve that characteristic rather than allowing the prediction to be interpreted as equivalent to a bounded forecast.

Open-Ended Prediction $\neq$ Time-Bounded Prediction

12. Relative Prediction Horizons

A Prediction Horizon may be defined relative to another identifiable event or state.

Examples include periods measured from:

prediction issuance; event occurrence; operational activation; system state transition; another governed temporal reference.

The reference point must be sufficiently identifiable for the horizon to remain reconstructable.

Relative Horizon $\neq$ Undefined Horizon

13. Recurring Prediction Horizons

Recurring predictive processes may apply to repeated horizons.

Each recurring horizon must remain sufficiently identifiable so that predictions produced for different periods are not silently treated as the same predictive instance.

Recurring methodology does not eliminate the need for horizon differentiation.

Recurring Prediction Process $\neq$ Single Continuous Prediction

14. Temporal Granularity

The Prediction Horizon must use temporal granularity appropriate to the Prediction Target and Predictive Purpose.

Seconds, minutes, hours, days, months, years, operational cycles, or other temporal units may carry materially different predictive meaning.

Excessively broad temporal granularity may make a target difficult to evaluate.

Excessively narrow granularity may imply unsupported predictive precision.

Temporal Granularity $\neq$ Predictive Accuracy

15. Target–Horizon Coherence

The Prediction Target and Prediction Horizon must form a coherent predictive proposition.

The same target may represent materially different predictions across different horizons.

For example:

Target X within one hour

and

Target X within ten years

must not be interpreted as equivalent merely because the target is identical.

Same Target + Different Horizon $\neq$ Same Prediction

PHDM preserves this distinction before probability or feasibility are subsequently determined.

16. Purpose–Horizon Coherence

The Prediction Horizon must remain compatible with the defined Predictive Purpose.

A horizon appropriate for long-term strategic planning may not be appropriate for immediate operational execution.

Likewise, a short-horizon prediction must not automatically be represented as supporting a long-term predictive purpose.

PHDM identifies temporal compatibility without selecting the predictive methodology.

Available Horizon $\neq$ Purpose-Appropriate Horizon

17. Horizon Ambiguity

PHDM must identify temporal ambiguity capable of materially changing the meaning of a prediction.

Material ambiguity may include:

unspecified start; unspecified end; unclear time zone; unclear interval interpretation; ambiguous relative timing; unclear recurrence; conflation of publication time and prediction horizon; unclear temporal units; multiple incompatible horizon interpretations.

A materially ambiguous horizon must not be treated as fully defined.

18. Horizon Differentiation

Predictions concerning the same object and target but applying to materially different horizons must remain distinguishable.

This supports later comparison of:

predictive methodology; probability; uncertainty; feasibility; revision behavior; outcomes; performance.

Horizon differentiation prevents predictive results from being compared without accounting for their temporal frame.

19. Horizon Change

A Prediction Horizon may require legitimate change.

A material horizon change must remain explicitly distinguishable from the original Prediction Horizon.

Extending a prediction after its original horizon has expired must not silently convert the original prediction into a successful longer-horizon prediction.

Horizon Extension $\neq$ Original Prediction Continuity

Likewise:

Missed Horizon + Later Outcome $\neq$ Original Prediction Success

unless the original predictive proposition explicitly governed that later outcome.

20. Retrospective Horizon Drift

PHDM must protect against retrospective alteration of the temporal frame after the outcome becomes known.

A prediction made for one period must not later be represented as though it applied to another period merely because the predicted outcome eventually occurred.

This distinction is essential for legitimate outcome association and predictive performance evaluation.

Eventual Occurrence $\neq$ Correct Prediction Within the Original Horizon

21. Horizon Continuity

The original Prediction Horizon must remain sufficiently identifiable throughout the Prediction Lifecycle.

Later revisions, extensions, contractions, or replacements must remain distinguishable from the original temporal proposition.

PHDM establishes the horizon reference that subsequent monitoring, revision, outcome, performance, and record governance can preserve.

22. Horizon–Feasibility Separation

Prediction Horizon may materially affect feasibility, but PHDM does not determine feasibility.

An outcome infeasible within one horizon may become feasible within another.

That determination belongs to PFES — Prediction Feasibility Standard .

PHDM supplies the temporal condition against which feasibility can subsequently be evaluated.

Prediction Horizon $\neq$ Prediction Feasibility

23. Horizon–Probability Separation

Probability may vary materially across Prediction Horizons.

PHDM does not assign or calculate probability.

It establishes the temporal frame to which a later probability representation must correspond.

Probability Without Horizon Context May Misrepresent Predictive Meaning

Probability governance belongs to PPUS — Prediction Probability & Uncertainty Standard .

24. Horizon Definition Sufficiency

A Prediction Horizon is sufficiently defined when it:

establishes the applicable temporal frame; identifies start and end conditions where relevant; distinguishes point, interval, recurring, relative, open, or bounded structure where material; uses sufficient temporal granularity; remains coherent with the Prediction Target; remains compatible with the Predictive Purpose; resolves material temporal ambiguity; can remain historically identifiable; supports subsequent boundary, methodology, feasibility, monitoring, outcome, and performance governance.

Horizon sufficiency does not establish prediction validity, feasibility, probability, or correctness.

25. Governance Requirements

PHDM requires that:

1. the applicable Prediction Horizon be identifiable where time is material;
2. the horizon remain associated with the correct Prediction Object and Target;
3. start and end references be established where applicable;
4. temporal structure be distinguishable where materially relevant;
5. temporal granularity be sufficient without artificial precision;
6. material temporal ambiguity be resolved;
7. target–horizon coherence be established;
8. purpose–horizon coherence be established;
9. materially different horizons remain distinguishable;
10. material horizon changes remain historically traceable;
11. eventual outcome occurrence not retrospectively redefine the original horizon.

26. Governance Logic

PHDM applies the following operational logic:

Identified Prediction Object

Defined Predictive Purpose

Specified Prediction Target

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Candidate Prediction Horizon

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Temporal Reference Identification

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Horizon Structure Determination

↓

Start / End Establishment where applicable

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Temporal Granularity Assessment

↓

Target–Horizon Coherence

↓

Purpose–Horizon Coherence

↓

Temporal Ambiguity Assessment

↓

Horizon Definition Sufficiency

↓

Defined Prediction Horizon

↓

PBEM — Predictive Boundary Establishment Module

Where horizon sufficiency is not established:

Horizon Gap → Clarification / Temporal Differentiation → Reassessment

27. Minimum Implementation Framework

A minimum implementation of PHDM must establish:

A. Prediction Object Reference

The Prediction Object to which the horizon applies.

B. Prediction Target Reference

The specified Prediction Target.

C. Predictive Purpose Reference

The defined Predictive Purpose.

D. Horizon Statement

A clear representation of the applicable temporal frame.

E. Horizon Type

Where applicable:

Point / Interval / Relative / Recurring / Bounded / Open

F. Horizon Start

The applicable starting point or condition where relevant.

G. Horizon End

The applicable endpoint or condition where relevant.

H. Temporal Granularity

The temporal resolution necessary to preserve predictive meaning.

I. Temporal Reference

Applicable clock, calendar, event, operational cycle, or other temporal reference where necessary.

J. Horizon Coherence State

Whether the horizon remains coherent with the target and purpose.

K. Horizon State

At minimum:

Undefined → Candidate → Clarification Required → Defined → Changed → Expired

Only a sufficiently Defined Prediction Horizon may proceed as the established horizon into final POPS boundary governance.

28. Governance Outputs

PHDM may produce:

Defined Prediction Horizon; horizon type; horizon start; horizon end; applicable interval; temporal granularity; temporal reference; target–horizon coherence state; purpose–horizon coherence state; horizon ambiguity state; horizon lifecycle state; unresolved temporal requirements.

The primary governed output is:

Defined Prediction Horizon

This proceeds with the established Prediction Object, Predictive Purpose, and Prediction Target into:

PBEM — Predictive Boundary Establishment Module

29. Critical Distinctions

Prediction Target $\neq$ Prediction Horizon

Prediction Creation Time $\neq$ Prediction Horizon Start

Prediction Publication Time $\neq$ Prediction Horizon

Prediction Horizon End $\neq$ Outcome Determination

Point Prediction Horizon $\neq$ Interval Prediction Horizon

Open-Ended Prediction $\neq$ Time-Bounded Prediction

Relative Horizon $\neq$ Undefined Horizon

Recurring Prediction Process $\neq$ Single Continuous Prediction

Temporal Specificity $\neq$ Artificial Temporal Precision

Temporal Granularity $\neq$ Predictive Accuracy

Same Target + Different Horizon $\neq$ Same Prediction

Horizon Extension $\neq$ Original Prediction Continuity

Missed Horizon + Later Outcome $\neq$ Original Prediction Success

Eventual Occurrence $\neq$ Correct Prediction Within the Original Horizon

Prediction Horizon $\neq$ Prediction Feasibility

30. Enterprise Application

PHDM provides a common temporal-governance layer across forecasting systems, AI models, prediction markets, enterprise planning, risk forecasting, financial prediction, weather forecasting, autonomous systems, infrastructure forecasting, and other predictive environments.

It allows organizations and machines to distinguish predictions that concern the same object and target but apply to materially different periods.

This supports legitimate probability representation, feasibility assessment, monitoring, revision, outcome association, calibration, performance comparison, and historical reconstruction.

31. Use Case 1 — Electricity Demand Forecast

An electricity operator predicts that demand will reach a specified level.

The Prediction Object and Target may remain identical across several forecasts, while the horizons differ:

next hour

tomorrow evening

next winter

within five years

PHDM establishes the temporal frame belonging to each predictive proposition.

A target reached five years later cannot retrospectively establish the correctness of a prediction whose governed horizon ended tomorrow evening.

32. Use Case 2 — Asset Price Prediction

An analyst predicts that an asset will reach a specified price.

Without a defined Prediction Horizon, the claim may remain impossible to evaluate meaningfully.

PHDM distinguishes:

reaches the price within 30 days

from:

reaches the price within five years

and from:

may reach the price at some unspecified future point.

The target may be identical, but the governed predictions are not.

33. Canonical Boundary

PHDM governs the identification, specification, structure, temporal reference, granularity, coherence, change, and continuity of the Prediction Horizon. It does not govern Predictive Boundary establishment, methodology, Prediction Basis, assumptions, prediction formation, probability, uncertainty, feasibility, collective prediction, communication, reliance, revision, Operational Reality determination, predictive performance, or Prediction Record continuity.

PHDM establishes when the Prediction Target applies .

It does not determine whether, when, or how the predicted outcome will actually occur .

Canonical Closing Statement

A prediction that eventually becomes true is not necessarily a prediction that was correct within the horizon in which it was made.

PHDM preserves the temporal meaning of a prediction so that later probability, feasibility, revision, outcome, and performance remain anchored to the time frame originally governed.

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